

Sankha Singhamahapatra

Kolkata, India

+91 7477362785 — mahapatrasamkh8@gmail.com — LinkedIn — GitHub



Professional Summary

Electronics and Communication Engineering undergraduate with experience in AI, embedded systems, renewable energy, and research-driven engineering projects. Skilled in machine learning, IoT systems, computer vision, and software development with practical exposure through internships, hackathons, and international design registration work.

Education

Techno Main Salt Lake B.Tech in Electronics and Communication Engineering	2023–2027
Bankura Christian Collegiate School Higher Secondary (10+2)	2021–2023

Technical Skills

Programming	Java, Python, JavaScript
Web Development	HTML, CSS, PHP, MySQL, XAMPP
Tools	GitHub, VS Code, MATLAB, Xilinx
Core Subjects	Machine Learning, DBMS, Computer Networks, Signal Processing, Digital Logic Design
Languages	Bengali, Hindi, English

Experience

Project Intern — Variable Energy Cyclotron Centre (VECC) Department of Atomic Energy, Government of India	2026
- Performed time-series analysis on accelerator datasets using machine learning models. - Worked on anomaly detection, model evaluation, and technical documentation. - Gained practical exposure to experimental nuclear physics research systems.	
Campus Ambassador — Physics Wallah	
Promoted technical initiatives and increased student engagement through outreach activities.	

Projects

Smart Solar Tracking System	ESP32, MATLAB, IoT
- Developed an intelligent solar tracking system using LDR sensors and servo motors. - Implemented real-time alignment algorithms for photovoltaic optimization.	
AI-Based Brain Tumor Detection	Python, CNN
Built a deep learning model for MRI-based brain tumor classification.	
Space Cargo Stowage System	PHP, MySQL
Designed a database-driven system for optimized spacecraft cargo arrangement.	

Patent & Design Registration

UK Design Registration — Intellectual Property Office (UK)	2026
- Registered design for Photovoltaic Panel Solar Tracking Device under the UK Registered Designs Act 1949. - Design Number: 6514788 - Contributed to intelligent tracking logic, sensor integration, and photovoltaic optimization mechanisms.	

Achievements & Certifications

- GATE ECE 2026 — Qualified (AIR - 7331) - National Space Hackathon 2025 — Winner (AIR 1), IIT Delhi & ISRO - Smart India Hackathon 2024 — Grand Finalist - Scientific Observations from Space — Indian Institute of Remote Sensing (IIRS), ISRO - NPTEL — The Joy of Computing Using Python (Elite), IIT Madras	
--	--